

Keshav Rajput

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Machine learning engineer, new grad. MS Computer Science, NYU Tandon, May 2026. PyTorch, LLM agents, Kaggle.

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EDUCATION

New York University, Tandon School of Engineering

Master of Science (MS) in Computer Science

Sep 2024 - May 2026

GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi

Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics

2020 - 2024

CGPA 8.19/10

Coursework: Machine Learning, Neuroinformatics, Foundations of Data Science, Advanced Machine Learning, Convex Optimization

PROJECTS

LFP2Vec audit | Python, PyTorch | github.com/happycoder/lfp2vec-audit

2026

- Reproduced LFP2Vec (NeurIPS 2025) fine-tuning on public IBL and Allen Neuropixels data on a laptop: 0.74 +/- 0.07 balanced accuracy vs 0.68 published.
- Traced the cross-lab collapse to a preprocessing mismatch (spectra diverge 768x above 300 Hz); a 100 Hz low-pass restores the published margin.
- Calibration under lab shift (ECE 0.10 in lab, 0.56 across, 0.35 after the fix) and an electrode-position control; 337 tests and leakage checks in CI.

Gemma 4 SWE agent (in progress) | Python, Google ADK

2026

github.com/happycoder/gemma-swe-agent

- Kaggle Gemma 4 Developer Agent (solo, in progress): an LLM agent in declarative Google ADK that fixes GitHub issues with Gemma 4 31B; final ranking pending.
- swelite, a re-implementation of the organizers' unreleased evaluation harness (Docker sandboxes, 9 tools, ADK YAML compiler): 109 of 129 public tasks pass gold checks.
- A 16 GB laptop GPU over SSH serves the vLLM proxy; found ADK never compacts context mid-task; temperature 0.7 cut identical retries from 13.5 to 5.1.

Biohub cell tracking (in progress) | Python, PyTorch, NumPy, SciPy, polars

2026

Builds on Igor Zharov's public notebook (Apache-2.0); the harness, classifiers and experiments are mine.

- Kaggle Biohub Cell Tracking: public leaderboard 0.952 against 0.947 for the public notebook it builds on, on free Kaggle GPU; final ranking pending.
- Offline harness runs GPU inference once, then re-tunes post-processing locally against the official scorer on 40 held-out videos, matching the notebook to +0.00011.
- 3D CNN trained on 134 annotated divisions prunes false forks: predicted +0.0047 on a pre-registered holdout, measured +0.005 on the leaderboard twice.

Blockwave | Python, numpy, pygame-ce, PyTorch, PPO | github.com/happycoder/blockwave

2026

- Falling-block game in Python and numpy with no asset files: SRS rotation, 7-bag, T-spins, 240 Hz fixed timestep, 300k engine steps/s headless.
- PPO agent on an empowerment reward that never sees score, lines or level, enforced by a static test; the pixel version sees an 88x88 crop.
- 465 tests in 40 s; after two identical 150M-step runs differed by 42%, withdrew every single-seed comparison.

COMPETITIONS

- **Kaggle Playground S6E9**, 2026. Out-of-fold AUC 0.9465 from nested target encodings, a LightGBM, CatBoost and MLP blend and 192 experiments judged by paired DeLong tests. Final ranking pending, ends 2026-09-30.
- **TrafficFlowBench**, 2026. Public leaderboard 0.845 against a 0.553 baseline, with LightGBM time-series queue forecasting and an NNLS origin-destination solver. In progress, ends 2026-11-06.
- **AI x GTM Hackathon (Block Convey), Revenue Intelligence track**, 2026. Entry is Ledger, a pipeline-review agent that checks every claim against the CRM. Event on 2026-10-17.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

ML: PyTorch, deep learning, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, NumPy, pandas, polars

LLMs: LLM agents (Google ADK, MCP, tool calling), vLLM, Ollama, LiteLLM, LoRA/QLoRA fine-tuning (PEFT), llama.cpp/GGUF quantization

Tools: Git, CI/CD (GitHub Actions), Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale, Claude Code

LEADERSHIP AND ACTIVITIES

- Open source: ephys-mcp on PyPI and the MCP Registry (24 tools, 52 tests); six fixes with regression tests to hawk-py